Date:13.01.2021

2020-2021 FALL SEMESTER HYDRAULICS COURSE FINAL EXAM

Name: **Surname:** **Student ID:**

Question 4 (15):

In a rectangular channel (b=4 m and y=1.5m), velocity is 1.5 m/s. The model of this channel will be contructed in a laboratory.
a) Which type of model (similarity) should be used?
b) Determine the velocity, time, discharge and force scale ratios if scale factor, r_L is chosen as **1/1a.**
c) Determine the model dimensions and draw in a figure.
d) Fill the below tables.
(Note: **a** will be the last digit of your student ID number)

Scale Factor (r_L)	**1/1a**
Velocity scale factor	?
Time scale factor	?
Discharge scale factor	?
Force scale factor	?

	Prototype value	Model value
Velocity (m/s)	2.2	?
Velocity (m/s)	?	0.5
Discharge (m^3/s)	6	?